MIST 5400 · AGENTIC AI WEEK · VIBE CODING EXERCISE 2

From Churn Notebook to Live Website

In class, about 15 minutes, finish at home if needed · needs your completed modeling_workflow.ipynb

The point

Last week you built a churn model in `modeling_workflow.ipynb`. A notebook is for you. Nobody on the retention team will open it. Today you turn that notebook into something a manager can click: an interactive website that presents the findings and, **given a new customer, shows the model's prediction**.

Discussion: What other features would you like on the website?

(discuss with neighbors, 5 min)

1. **Must have: A live scorer.** A form with the customer's fields. Change a field and the churn probability updates. This is the "deploy the model" part.
2. **Must-have:** introduction of the website, of what it does.

Step 1 · Get the trained model ready

Download your completed `modeling_workflow.ipynb` (The one you uploaded to eLc for the modeling_workflow in-class exercise.) and drag it to a GenAI tool. Go to ChatGPT, Claude, Gemini, or Lovable, Google AI Studio.

Step 2 · Vibe code the site

The specifics are what make it work. Based on what you discussed, ask for that feature in the prompt. Here's an example prompt.

> **BUILD PROMPT**
> I trained a logistic regression to predict telecom customer churn. Build a single file HTML website that presents the findings to a retention manager. Include: _____. a model evaluation section with a confusion matrix that updates when I drag a threshold slider, and a form where I enter a new customer's fields and the page computes the churn probability in the browser using the coefficients.
> Make it look professional. No external servers. Use the actual numbers from the trained model.

Then steer. Typical second and third prompts:

- "Explain in one sentence what the threshold does."

Step 3 · Verify the deployment (Skip in-class so save time. But, this is important!)

We'd take one actual customer in the test set, run the notebook to get the predicted probability. Feed the customer features into the website and check if the shown prediction is the same as our manually-run output.

Submit

Under Assignments in **eLC -> Assignment -> Vibe-coding Churn Site**: your HTML file.